

3DV 2027 Submission #524 — Rebuttal / Camera-Ready Preparation

Anticipated challenges and prepared responses (post-deadline evidence)

Internal working document — not for submission as-is

September 2, 2026

Purpose

The submission and its supplementary are frozen. This document collects, for each anticipated reviewer challenge, the prepared response and the post-deadline evidence backing it. Supplementary Version B (PATHA_supp_reproducibility_vB.pdf) holds the camera-ready-quality write-up of items 1–2; this document is the playbook.

1 The learning-rate challenge (highest risk)

Challenge. “You train the baseline at RESfM’s released 10^{-3} but your arms at 10^{-4} . Retraining the supervised baseline at 10^{-4} may close your out-of-distribution margins.”

What a reviewer would find. It is true and we have measured it: the baseline retrained at 10^{-4} improves on contaminated OOD — 1DSfM 5.90 (vs. 11.03 declared), 1DSfM-hard 12.81 (vs. 16.22), BlendedMVS 0.14 (vs. 0.38) — while degrading in-distribution (0.53 vs. 0.34) and on Strecha (1.73 vs. 0.006).

Prepared response (four points, all measured).

1. The policy was disclosed, not hidden: the paper’s Protocol section states the per-method base learning rate and the convergence-timing rationale; neither method is truncation-limited.
2. Not an under-training artifact: a 30k-epoch replicate at 10^{-4} plateaus mid-schedule (best epoch 16k; never improves through 29,999), so the OOD gain is a regularization effect of the lower rate, not unfinished training. Its full row: MegaDepth 0.36, 1DSfM 6.12, hard 13.88, Strecha 1.64, BMVS 0.18, Olsson 8.27.
3. The asymmetry runs against us, not for us: the symmetric change on our side (label-free arms at 10^{-3}) helps nowhere — the per-method policy is each method at its own best-converging rate, not a knob tuned in our favor.
4. The headline clean-benchmark findings are lr-robust: the Olsson collapse persists at both rates (8.3–8.5) and Strecha degrades at 10^{-4} ; and even against the strongest supervised variant (the 30k/ 10^{-4} row), label-free plain removal still leads at 60% contamination (12.4 ± 2.1 vs. 13.9).

Concession to offer if pressed: scope the 43% removal-family margin as “vs. the released-configuration baseline”; the removal medians (1.68 ± 0.19) remain ahead of the 10^{-4} baseline’s 1DSfM medians as well.

2 The single-seed challenge

Challenge. “All numbers are seed 20; the conclusions may be seed luck.”

Prepared response. Post-deadline we retrained every arm and the baseline under five seeds (20–24) and re-ran the full evaluation (36 cells, plus a 2,352-evaluation paired replication where the evaluation seed equals the training seed). Results (Version B, Sec. E):

- 60% contamination: worst removal seed 15.8 < best baseline seed 16.2 (every pairing).

- 43%: every removal median (1.4–1.9) beats every baseline median (2.4–4.3).
- Olsson collapse: 4/5 scratch seeds and 4/5 authors-faithful retrains (8.66–9.02; one escape at 2.32, mirroring the scratch escape at 2.69), plus the released checkpoint (9.10); every mad/weight/remove arm seed lands at 2.9–3.6.
- BlendedMVS: mad/weight arms 0.14 ± 0.02 below every baseline seed (0.35 ± 0.04).
- Eval-seed control: released weights under five evaluation seeds move by ± 0.008 ; the paired replication reproduces every band within a fraction of its std. The variance is in training, amplified by BA — not in our evaluation.
- Prior work in this line reports single-seed results (RESfm: “We fix the seed to 20”); we are, to our knowledge, the first in this benchmark family to report full training-seed bands.

3 The baseline-fidelity challenge

Challenge. “Your RESfm-from-scratch (0.344) is weaker than the released checkpoint (0.209) and the paper’s 0.169 — maybe your baseline is just badly trained.”

Prepared response. (a) The frozen supplementary already traces the released-vs-paper gap to environment-level BA basin selection (bidirectional, scene-concentrated flips; 17/36 scenes better in our runs); differences $\lesssim 0.05$ are environment noise. (b) Post-deadline, five authors-faithful retrains (authors’ exact validation metric and margin, seeds 20–24) give a MegaDepth band of 0.32 ± 0.11 — statistically indistinguishable from our declared scratch band 0.37 ± 0.12 — and reproduce the scratch band on every dataset, including the same OOD failures (Olsson collapse in 4/5 seeds, 8.66–9.02; BMVS 0.35 ± 0.03) and even the one-bad-Strecha-seed pattern. The two recipes draw from the same distribution; our baseline is not a weak retrain.

4 The TTT-nondeterminism challenge

Challenge. “Your weight+TTT numbers vary run to run; how much of the paper rests on that?”

Prepared response. Quantified in the frozen supplementary (± 0.07 on TTT means; deterministic no-TTT cells reproduce exactly). None of the headline claims (60%, 43%, Olsson, BlendedMVS) rests on a TTT cell; the paired five-seed program confirms the TTT cells are the only ones that move (~ 0.13) under evaluation re-seeding.

5 Camera-ready plan

1. Replace main-paper Tables 2, 5, 6, 7 with the extended versions (rotation, N_r , N_c) already staged in the supplementary (Sec. J), translation cells upgraded to five-seed bands.
2. Fold Version B’s Sec. E (five-seed bands, variance source, lr sensitivity) into the supplementary proper; retire the single-seed limitation.
3. Correct “every seed” phrasing for the Olsson collapse to the 4/5-plus-references statement; add the faithful-retrain replication row.
4. Disclose the 10^{-4} baseline row as a reference row with the four-point response of Sec. 1 condensed to a footnote-length remark.